Pavan Kumar Sathya Venkatesh

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Education

University of Massachusetts, Amherst **2025 - Present**
MS in Computer Science *3.90/4.00 GPA*

Vellore Institute of Technology, Chennai **2021 - 2025**
B.Tech in Computer Science & Engineering *8.91/10 CGPA*

Research Projects

Ongoing Projects

- Venkatraman, S.*, Raj, R.*, **Kumar, P. S.*** (2025). "PCM-NeRF: Probabilistic Camera Modeling for Neural Radiance Fields under Pose Uncertainty."
- Venkatraman, S.*, **Kumar, P. S.***, Ramachandran, V., Elias, S. (2025). "EndoBuddy: AI-Assisted Real Time Landmark Detection for Upper Gastrointestinal Endoscopy
- Venkatraman, S.*, Raj, R.*, **Kumar, P. S.*** (2025). "TIDE: Two-Stage Inverse Degradation Estimation with Guided Prior Disentanglement for Underwater Image Restoration".

Projects Under Review/Published

- Venkatraman, S.*, **Kumar, P. S.***, Raj, R.*, Chandrakala S (2025). "UGPL: Uncertainty-Guided Progressive Learning for Evidence-Based Classification in Computed Tomography." Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops.
- Venkatraman, S., **Kumar, P. S.**, Pandiyaraju, V., Abeshek, A., Aravintakshan, S. A., Kannan, A. (2025). "SPROUT: Symptom-centric Prototypical Representation Optimization and Uncertainty-aware Training for Few-Shot Precision Agriculture." NeuroComputing.
- **Kumar, P. S.**, Kumar, M. S., Thayyil, P. J., Pandiyaraju, V. (2025) Making Lies Visible: CM-TGT for Graph-Based Cross-Modal Deception Detection (Submitted: IEEE Transactions on Affective Computing)
- Venkatraman, S., Pandiyaraju, V., Abeshek, A., **Kumar, P. S.**, Aravintakshan, S. A., Kannan, A (2025). "Bayesian Uncertainty Propagation for Bone Fracture Diagnosis via Region-Aware Adaptive Label Refinement." Computers in Biology and Medicine. (Under review).
- Venkatraman, S., Pandiyaraju, V., Abeshek, A., **Kumar, S. P.**, & Aravintakshan, S. A. (2025). "Leveraging Bi-Focal Perspectives and Granular Feature Integration for Accurate, Reliable Early Alzheimer's Detection." IEEE Access, 13, 28678-28692.
- Venkatraman, S., Pandiyaraju, V., Abeshek, A., Aravintakshan, S. A., **Kumar, P. S.**, & Madhan, S. (2024). "Targeted Neural Architectures in Multi-Objective Frameworks for Complete Glioma Characterization From Multimodal MRI." Applied Soft Computing. (Under review)
- Pandiyaraju, V., Venkatraman, S., **Kumar, P. S.**, Malarvannan, S., & Kannan, A. (2024). "Exploiting Precision Mapping and Component-Specific Feature Enhancement for Breast Cancer Segmentation and Identification." Ain Shams Engineering Journal. (Under review).
- Pandiyaraju, V., Senthil Kumar, A. M., Praveen, J. I. R., Venkatraman, S., **Kumar, S. P.**, Aravintakshan, S. A., Abeshek, A., & Kannan, A. (2024). "Improved Tomato Leaf Disease Classification Through Adaptive Ensemble Models With Exponential Moving Average Fusion and Enhanced Weighted Gradient Optimization." Frontiers in Plant Science, 15.

Work Experience

MedxAI Innovations Pvt Ltd. **AI/ML and Software Developer Intern**
Chennai, India *May 2024 - Jul 2024*

- **EndoBuddy**
  - Improved accuracy of landmark identification in Upper-GI Endoscopy by 30% by introducing EndoVision, a model quantization based algorithm along with attention mechanisms.
  - Developed an automated feedback system to assist gastroenterologists by analyzing time spent on each landmark during endoscopy, ensuring accurate and error-free procedures.
- **Mediscan**
  - Built an automated report generating application which could be used for different clinical procedures
  - Enhanced workflow efficiency by 60% and reduced data entry efforts by 70% in Anderson Diagnostics and Labs, streamlining processes and optimizing productivity.
- **CerviLens**
  - Leveraged Graph Neural Networks to develop a system based on Jetson Nano for improving cervical cancer-grade classification during colposcopy procedures.
  - Improved early-stage cancer lesion detection by integrating the system with multiscale lenses enhancing image resolution by 30%

Prodapt Solutions Pvt Ltd. **Backend Developer Intern at NextGen Labs, Department of Delivery**
Chennai, India *Sep 2023– Oct 2023*

- Conducted Time Series Analysis on the datasets provided by the company by utilizing machine learning techniques to identify trends, seasonality and anomalies.
- Utilized Flask API framework to develop application which involved authentication, session creation and database migration
- Developed an application for Optical Character Recognition (OCR) to extract specific text from documents and images.

Projects

ItinerEase (Smart India Hackathon 2024) **Aug 2024 - Sep 2024**

- Tech Stack: Python, Django, HTML, CSS, JavaScript, AJAX, GPT-4 API
- Developed a personalized itinerary generator application for tourists powered by AI
- Elevated user satisfaction by improving trip recommendations utilizing RAG and increased personalization by applying continual learning

AI-COP **Dec 2024 - Jan 2025**

- Tech Stack: Python, Django, HTML, CSS, JavaScript, AJAX
- Built a Speech-to-Text transcription application which records voice notes from physicians and transcribes it to text using LLMs appropriately .
- Improved workflow efficiency by 60% in Kauvery Group of Hospitals, allowing medical professionals to dictate report findings directly, bypassing manual transcription and accelerating the creation of comprehensive medical records.

Augmented Autonomous Vision using GANs **Mar 2024 - Jul 2024**

- Tech Stack: Pytorch
- Performed realistic day-to-night image translation and vice-versa by utilizing a deep convolutional CycleGAN-based solution for enhancing the adaptability and safety of autonomous vehicles in varying lighting conditions
- Boosted visual data robustness by optimizing image quality and environmental representation and improved AV's perception for safer navigation
- Earned access to AMD Radeon Instinct Cloud Accelerator with ROCm 6.1.2, worth $24,000.

Technical Skills

Languages: Python, C, C++, Java, SQL, HTML5, CSS3, JavaScript, R, Electron and PHP
Frameworks, Cloud and Databases: PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face, streamlit, OpenCV, NLTK, Django, Postman, Flask, AMD Accelerator, PostgreSQL and MongoDB
Undergraduate Coursework: Machine Learning, Artificial Intelligence, Natural Language Processing, Computer Networks, Cryptography and Network Security, Operating Systems, Theory of Computation, Calculus, Differential Equations and Transforms, Probability and Statistics, Complex Variables and Linear Algebra
Graduate Coursework: COMPSCI589-Machine Learning, COMPSCI532-Systems for Datascience, COMPSCI602-Research Methods for Empirical CS, COMPSCI692-Neural Networks, COMPSCI690S-AI Alignment, COMPSCI520-Software Engineering

Extracurricular Activities

- Part of the operations department of the University's Game Development Club and extended support in terms of marketing and event management
- Served as a Teaching Assistant at Machine Learning Lab (BCSE424P) in VIT Chennai
- Served as a part of the organizing team for a university-wide gaming tournament with over 300 participants.